L7-1a

Lecture 7

Linear Algebra Done Right

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Vector Space

Vector Space V is a set on which two operation vector addition (+) and scalar multiplication (·) are defined

Basic example is n-dimensional Euclidean space $\mathbb{R}^n$ where every element is represented by a list of n real numbers, scalars are real numbers, addition is component wise, and scalar multiplication is multiplication on each term separately.

n-space $\mathbb{R}^n$ is called real vector space.

$\mathbb{C}^n \rightarrow$ complex vector space

For general vector space, scalars are members of a field F; in which case V is called a vector space over F.

In order for V to be vector space, following conditions must hold for all elements $X, Y, Z \in V$ and any scalars $r, s \in F$

1) Commutativity

$$X + Y = Y + X$$

2) Associativity

$$(X + Y) + Z = X + (Y + Z)$$

3) Additive Identity

$$0 + X = X + 0 = X$$

4) Additive Inverse

$$X + (-X) = 0$$

5) Associativity of scalar multiplication

$$r(sX) = (rs)X$$

6) Distributivity of scalar sum

$$(r+s)X = rX + sX$$

7) Distributivity of vector sums.

$$r(X+Y) = rX + rY$$

8) Scalar multiplication identity

$$1X = X$$

Example

$V = \mathbb{R}^2$ is a vector space over field of real no.

$\alpha \rightarrow$ $(a,b), (c,d) \in \mathbb{R}^2$

$$(a,b) + (c,d) = (a+c, b+d) \in \mathbb{R}^2$$

$\therefore$ closure holds in $\mathbb{R}^2$

$\beta \rightarrow$ $(a,b) + (c,d) = (a+c, b+d) = (c+a, d+b)$

$$= (c,d) + (a,b)$$

$\therefore$ Commutativity holds in $\mathbb{R}^2$

L7-2a | $\gamma \Rightarrow (a,b),(c,d),(e,f) \in \mathbb{R}^2$.

Then $(a,b)+((c,d)+(e,f)) = \cancel{(a+c+e,\ b+d+f)}$

$$= (a,b)+(c+e,\ d+f)$$

$$= (a+(c+e),\ b+(d+f)) = ((a+c)+e,\ (b+d)+f)$$

$$= (a+c,\ b+d)+(e,f)$$

$$= ((a,b)+(c+d)) + (e,f)$$

Associative property holds in $\mathbb{R}^2$

$\delta \Rightarrow (a,b) \in \mathbb{R}^2$, Then $(0,0) \in \mathbb{R}^2$

$$(0,0)+(a,b) = (0+a,\ 0+b)$$

$$= (a,b)$$

$(0,0)$ is identity elements in $\mathbb{R}^2$

$\xi \Rightarrow (a,b) \in \mathbb{R}^2$; $(-a,-b) \in \mathbb{R}^2$

$$(a,b)+(-a,-b) = (a-a,\ b-b) = (0,0)$$

$\alpha, \beta, \gamma, \delta, \xi$ $(\mathbb{R}^2,+)$ Abelian Group

$\Rightarrow$ k is scalar $k \in \mathbb{R}$; $(a,b) \in \mathbb{R}^2$ Then

$$k(a,b) = (ka,\ kb) \in \mathbb{R}^2$$

$\Rightarrow$ $k \in \mathbb{R}$

$(a,b), (c,d) \in \mathbb{R}^2$

Then $k((a,b)+(c,d)) = k(a+c, b+d)$

$= (ka+kc, kb+kd)$

$= (ka,kb) + (kc,kd) = k(a,b) + k(c,d)$

$\Rightarrow$ $k, m \in \mathbb{R}$ $\quad (a,b) \in \mathbb{R}^2$

$(k+m)(a,b) = ((k+m)a, (k+m)b)$

$= (ka+ma, kb+mb)$

$= (ka,kb) + (ma,mb)$

$= k(a,b) + m(a,b)$

$\Rightarrow$ $k, m \in \mathbb{R}$ $\quad (a,b) \in \mathbb{R}^2$

$k(m(a,b)) = k(ma, mb)$

$= (k(ma), k(mb)) = ((km)a, (km)b)$

$(km)(a,b)$

$\Rightarrow$ $(a,b) \in \mathbb{R}^2$

$1(a,b) = (1a, 1b) = (a,b)$

$(\mathbb{R}^2, +, \cdot)$ is a vector space

Same can be done for $\mathbb{R}^2$ and so for $\mathbb{R}^N$

L 7 - 3a

(1) Let $V = M_{m\times n}$ is a set of matrices of order $m \times n$. $M_{m\times n}$ forms a vector space over $\mathbb{R}$.

Vector addition is just matrix addition & scalar multiplication is defined in the obvious way.

(2) $V = \mathbb{R}^n$ forms a vector space over $\mathbb{R}$.

(3) $V = P$ forms vector space over $\mathbb{R}$ where P = set of all polynomials in variable x and coefficient from $\mathbb{R}$

(4) $V = P_n$ forms a vector space over $\mathbb{R}$ where P_n = set of all polynomials of degree atmost n in variable x and coefficient from $\mathbb{R}$.

V= set of all polynomial of degree n doesn't form a vector space over $\mathbb{R}$

$v_1 = 9x^4 + 6x^3 + 3x^2 + x + 7 \in V$

$v_2 = -9x^4 + 6x^3 + 4x^2 + 3x + 1 \in V$

$v_1 + v_2 = 12x^3 + 7x^2 + 4x + 8 \notin V$ (as poly of degree n)

Closure property doesn't hold.

Geometric interpretation of $\mathbb{R}^n$

$n = 1, 2$ or 3;

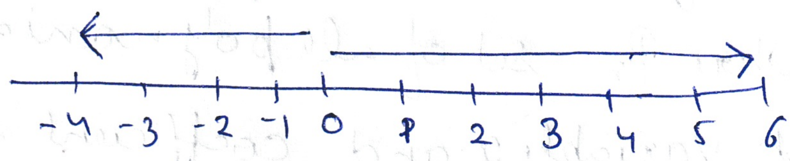
Space $\mathbb{R}^n$ has a very useful geometric interpretation

$n = 1$,

$v = \mathbb{R}$

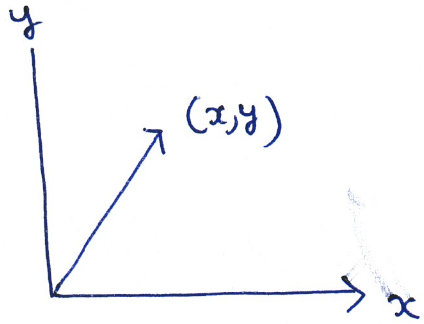
$v = 6$,

$v = -4$



$n = 2$ $v = \mathbb{R}^2$

$v = (x, y)$



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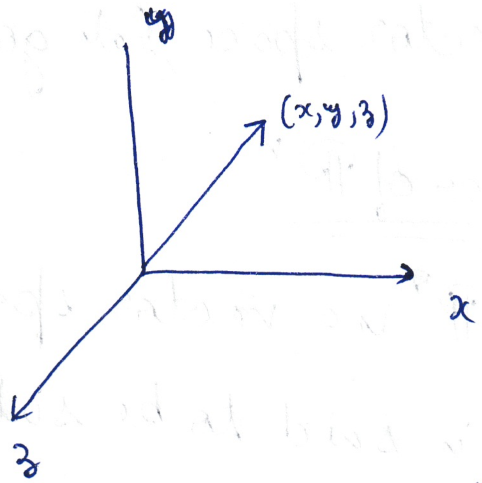
$n=3$

$V=\mathbb{R}^3$

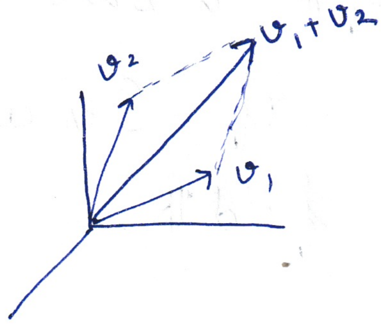
$v=(x,y,z)$



v_1, v_2 in $\mathbb{R}^3$

$v_1=(x_1,y_1,z_1)$

$v_2=(x_2,y_2,z_2)$



$v_1+v_2=(x_1+x_2, y_1+y_2, z_1+z_2) \in \mathbb{R}^3$

If v_1, v_2 and v_3 are linearly independent, which mean v_1 & v_2 are no collinear. and vector v_3 doesn't lie in plane of v_1 and v_2

Euclidean Space

Vectors in Euclidean space consists of n-tuples of realnumber.

$$x=(x_1, x_2, \ldots, x_n)$$

Euclidean space is used to represent physical space, with distances, length & angles.

For $n=3$; it is easy to understand but beyond $n>3$; it is hard to visualize.

After vector space, lets go to subspaces.

Subspace of $\mathbb{R}^n$

If $\mathbb{R}^n$ is a vector space over $\mathbb{R}$, then a subset S of $\mathbb{R}^n$ is said to be subspace of vector space $\mathbb{R}^n$ if S is also a vector space over the same field $\mathbb{R}$.

$\mathbb{R}^n \to$ vector space

S is non-empty

$\boxed{\subseteq \text{ - subset of}}$

$S \subseteq \mathbb{R}^n$ is said to be subspace of $V = \mathbb{R}^n(\mathbb{R})$ if

a) $0 \in S$ [should contain zero vector]

b) S is closed under vector addition

$x, y \in S \Rightarrow x + y \in S$

c) S is closed under scalar multiplication

$x \in S, \alpha \in \mathbb{R} \Rightarrow \alpha x \in S$

S is the set of all ordered pair (x_1, x_2, x_3) belongs to $\mathbb{R}^3$ such that $x_1 = x_2 = x_3$

Example

$| \to$ such that

$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = x_2 = x_3\}$ is a subspace of $\mathbb{R}^3$

$v_1 = (x_1, x_2, x_3)$ & $v_2 = (y_1, y_2, y_3)$ are 2 vectors in S, then

$av_1 + bv_2 =$

$(ax_1 + by_1, ax_2 + by_2, ax_3 + by_3)$

$x_1 = x_2 = x_3 \mid y_1 = y_2 = y_3 \mid ax_1 = ax_2 = ax_3$, $by_1 = by_2 = by_3 \mid \Rightarrow ax_1 + by_1 = ax_2 + by_2 = ax_3 + by_3$

L7-5a

b) $S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 - x_3 = 5\}$

$v_1 = (x_1, x_2, x_3)$; $v_2 = (y_1, y_2, y_2)$

~~$v_1 + v_2$~~

$0 + 0 + 0 = 5$ Not possible

S is not a subset.

c) $S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 - x_3 = 0\}$

$(0,0,0)$ $\quad 0 + 0 - 0 = 0$ ✓

$av_1 + bv_2$

$(ax_1 + by_1, ax_2 + by_2 + ax_3 + by_3)$

$ax_1 + by_1 + ax_2 + by_2 - ax_3 - by_3$

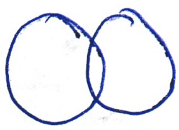
$$\underbrace{a(x_1 + x_2 - x_3)}_{=0} + \underbrace{b(y_1 + y_2 - y_3)}_{=0} = 0$$

also element of S.

S is subspace of $\mathbb{R}^3$

Note

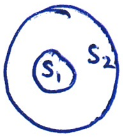
☆ The intersection of any non-empty collection of subspace of $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$



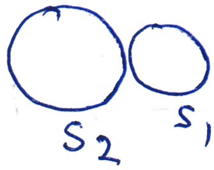
$S_1 \cap S_2$ is subspace of V.

$S_1 \quad S_2$

★ The union of two subspace of $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ if & only if one of them is contained in other



$S_1 \cup S_2$ subspace of V



$S_1 \cup S_2$ not subspace of V

Span → set of all vectors obtainable by a linear combination of original vectors

Linear span

Let $S = \{v_1, v_2, \dots, v_n\}$ be a non empty subset of V

V is a vector space

Then one can have a set

$$L(S) = \{a_1 v_1 + a_2 v_2 + a_3 v_3 + \dots + a_n v_n \mid a_i \in \mathbb{R}, 1 \le i \le n\}$$

is a linear span of set S.

$$\mathbb{R}^2 = L(\{(0,1), (1,0)\})$$

$$L(s) = a_1(0,1) + a_2(1,0) = (a_1, a_2) \quad a_1, a_2 \in \mathbb{R}$$

$$\mathbb{R}^3 = L(\{(1,0,0), (0,1,0), (0,0,1)\})$$

$$L(s) = a_1(1,0,0) + a_2(0,1,0) + a_3(0,0,1)$$
$$= (a_1, a_2, a_3)$$

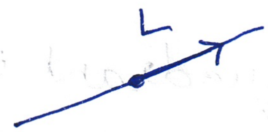
L7-6a

Result ① Let $V(F)$ ~~be~~ is a vector space and let S be a non-empty subset of V. Then $L(S)$ is a subspace of V

② Let S be a non-empty subset of vector ~~subset~~ space V. Then $L(S)$ is smallest subspace of $V(F)$ containing S.

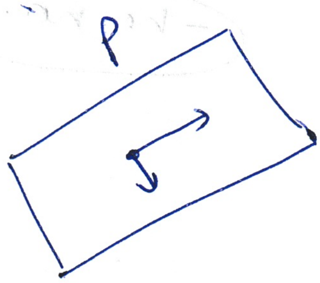
Example

① A Line through the origin



L contains zero, and is easily seen to be closed under addition & scalar multiplication

② A plane passing through origin



P contains zero.

Sum of two vectors in P is also in P.

Any scalar multiple will be in P.

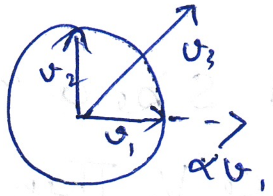
Non-example

(1) A circle.

A unit circle C is not a subspace.

Not closed under addition.

Not closed under multiplication



$v_1 + v_2 = \underline{v_3}$

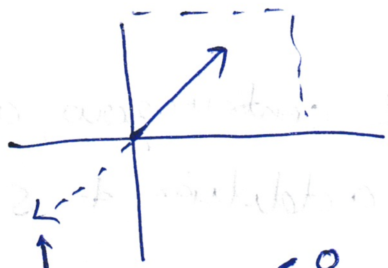
$\alpha v_1 = \alpha v_1 \to$ outside circle

(2) First quadrant in $\mathbb{R}^2$ is not a subspace

Contains origin ✓

Closed under addition ✓

However not closed under scalar multiplication.



-ve no. ←

α is scalar

L7-7a

$$A_{m,n} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{pmatrix}$$

where $a_{ij} \in \mathbb{R}$.

Then

1) Row space of A is given by $L\{(a_{11}, a_{12}, a_{13}, \ldots, a_{1n}), (a_{21}, a_{22}, a_{23}, \ldots, a_{2n}), \ldots, (a_{m1}, a_{m2}, a_{m3}, \ldots a_{mn})\}$ is a subspace of $\mathbb{R}^n(\mathbb{R})$

2) Column space of A is given by $L(\{(a_{11}, a_{21}, a_{31}, \ldots a_{m1}); (a_{12}, a_{22}, \ldots a_{m2}); \ldots; (a_{1n}, a_{2n}, \ldots a_{mn})\})$ is a subspace of $\mathbb{R}^m(\mathbb{R})$

3) Set $N(A) = \{x \in \mathbb{R}^n \mid AX = 0\}$ is said to be nullspace of A. The null space is defined to be the soln. set of the $AX = 0$.

4) Set $R(A) = \{b \in \mathbb{R}^n \mid AX = b\}$ for some $X \in \mathbb{R}^n\}$ is said to be range of A

In general column space$(A) = R(A)$

Basis

Let $V(F)$ be a vector space.

A basis for V is a set of linearly independent vectors in V which spans the vector space V.

→ The space V is finite-dimensional if it has a finite basis

→ A vector in basis is called basis vector

$B = \{v_1, v_2, v_3, \ldots, v_n\}$ is a basis of a n-dimensional vector space V if.

① Set of vector $\{v_1, v_2, v_3, \ldots, v_n)$ is linearly independent.

② For any vector $v \in V$, we have

$v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \ldots + \alpha_n v_n$

Where $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n$ are scalar from the field F

L7-8a

① $V = \mathbb{R}^3(\mathbb{R})$

possible basis are $\{(1,0,0),(0,1,0),(0,0,1)\}$

② $V = \mathbb{R}^n(\mathbb{R})$

$\{(1,0,0,\ldots 0),(0,1,\ldots 0),\ldots(0,0,0,\ldots,0,1)\}$

These are standard basis for $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^n$

③ $V = \{M_{2\times2}$ | set of all 2×2 real matrices

$\left\{\begin{pmatrix}1&0\\0&0\end{pmatrix},\begin{pmatrix}0&1\\0&0\end{pmatrix};\begin{pmatrix}0&0\\1&0\end{pmatrix};\begin{pmatrix}0&0\\0&1\end{pmatrix}\right\}$

linearly independent

4 vector

linear combination of 4 matrices.

④ $V = \{M_{2\times2}$ | set of all 2×2 real ~~(M)~~ symmetric matrices

~~There those are not b4~~

The $\left\{\begin{pmatrix}1&0\\0&0\end{pmatrix};\begin{pmatrix}0&1\\1&0\end{pmatrix};\begin{pmatrix}0&0\\0&1\end{pmatrix}\right\}$

This is symmetric matrix

3 vector

Dimension

The no. of vector/elements in the basis of $V(F)$ is called dimension of vector space $V(F)$

Ex

→ Dimension of vector space $\mathbb{R}^n(\mathbb{R})$ is n.
→ Dimension of vector space $M_{m \times n}(\mathbb{R})$ is $m \times n$.
→ Dimension of vector space $P_n(\mathbb{R})$ is $n+1$.
→ Dimension of vector space P (set of all Polynomials with coefficients from $\mathbb{R}$) is ∞.

Important Results

→ Let $(v_1, v_2, \ldots, v_n)$ be a basis of given vector space $V(F)$.
If S be a subset of vectors having more than n vectors, then set S is linearly dependent

→ Let $(v_1, v_2, \ldots, v_k)$ be a linearly independent subset of n-dimensional vector space $V(F)$, where $k < n$, Then S ~~is~~ can be extended to a basis of $V(F)$
I can have n linearly independent vector

L7-9a)

Lets say that $\mathbb{R}^7$

I have 4 linearly idendent vector of $\mathbb{R}^7$

then I can add 3 more l.i. vectors &

create basis of $\mathbb{R}^7$

★→ Let $V(F)$ be a finite-dimensional vector space. Then any two bases of V have the same no. of vectors. (dimension of vector space)

$\mathbb{R}^3(\mathbb{R})$

$B_1 = \{(1,0,0),(0,1,0),(0,0,1)\}$

$B_2 = \{(1,0,0),(1,1,0),(1,1,1)\}$

linearly independent set ↑

Many other such basis

★ Let $V(F)$ be a finite-dimensional vector space and let S_1 & S_2 be 2 subspaces of V. Then

$\dim(S_1) + \dim(S_2) = \dim(S_1+S_2) + \dim(S_1 \cap S_2)$

Example

① Find basis & dimension of subspace S of $\mathbb{R}^3$ given as

$S = \{(x_1, x_2, x_3) \mid x_1 + x_2 - x_3 = 0\}$

$\{(x_1, x_2, x_1 + x_2)\}$

$\{x_1(1,0,1) + x_2(0,1,1)\}$

Hence basis of S is $\{(1,0,1), (0,1,1)\}$

Therefore $\dim(S) = 2$.

② $V = \mathbb{R}^3(\mathbb{R})$

$W = \{(x_1, x_2, x_3) \mid x_1 = x_2 = x_3\}$

$= \{(x_1, x_1, x_1) \in \mathbb{R}^3\}$

Basis of W is $\{(1,1,1)x_1 \in \mathbb{R}\}$
$\{(1,1,1)\}$ $\quad \mathrm{Dim}(W) = 1$

Basis of $S \cap W$

$S \cap W \ \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 - x_3 = 0 \ \& \ x_1 = x_2 = x_3\}$

$= \{(x_1, x_2, x_1 + x_2) \mid x_1 = x_2 = x_3\}$

$\{(x_1, x_1, 2x_1)$ only possible of $(0,0,0)$

$B_{S \cap W} = \{(0,0,0)\}$ $\quad \mathrm{Dim}(S \cap W) = 0$

↑ empty basis.

L7-10a

~~Linear Transformation~~

The concept of basis is really important for "dimensional reduction"

Linear Transformation

↓ useful for 2D or 3D animation or modeling

Object size or shape needs to be trans-formed from one viewing angle to the next.

Object can be rotated and scaled. within space using linear transformation.

→ One can also use linear transformation to project the data from one space to another

Useful in Classifier → as some time dataset is not separable in original space but after transformation one can separate in another space with different dimensions

L7-10b

linear transformation

Let V and W be vectors space over a field F of dimension n and m, respectively.

A linear transformation is a mapping T:

$$V^{(n)}(F) \to W^{(m)}(F)$$ such that

a) $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all v_1 & $v_2 \in V$

b) $T(\alpha v) = \alpha T(v)$ for all $v \in V$ & $\alpha \in F$.

α is scalar.

Simple Example

$T : \mathbb{R}^3 \to \mathbb{R}^3$ such that

$$T(x_1, x_2, x_3) = (x_2, x_1, 0)$$

$$T(v_1) = T(x_1, x_2, x_3) = (x_2, x_1, 0)$$

$$T(v_2) = T(y_1, y_2, y_3) = (y_2, y_1, 0)$$

$$T(v_1 + v_2) = T(x_1+y_1, x_2+y_2, x_3+y_3) = (x_2+y_2, x_1+y_1, 0)$$

$$= (x_2, x_1, 0) + (y_2, y_1, 0)$$

$$T(v_1) + T(v_2)$$

$$\alpha T(\alpha v) = T(\alpha x_1, \alpha x_2, \alpha x_3) = (\alpha x_2, \alpha x_1, 0)$$

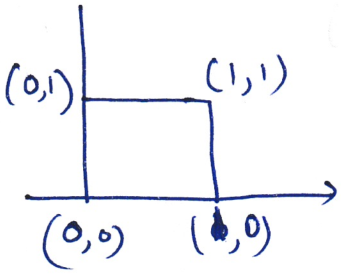
$$= \alpha(x_2, x_1, 0)$$

$$\alpha T(v)$$

Hence T is a linear transformation

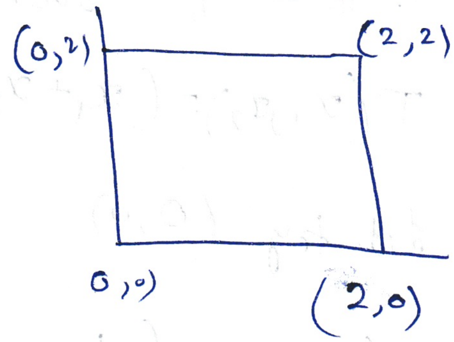
L7-11a | Linear transformations

when applied - change, rote

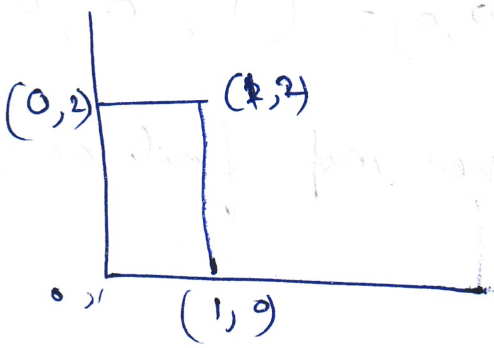


$T : \mathbb{R}^2 \to \mathbb{R}^2$

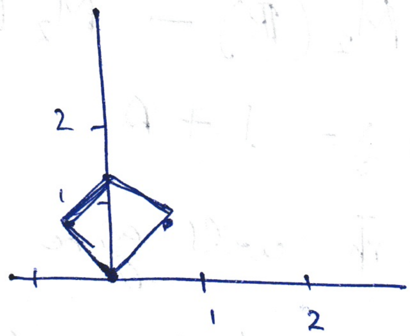
$T(x_1, x_2) = (2x_1, 2x_2)$



$T(x_1, x_2) = (x_1, 2x_2)$



rotate



$T(x_1, x_2) =$

$(x_1 \cos\theta - x_2 \sin\theta, x_1 \sin\theta + x_2 \cos\theta)$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\theta = 0$ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\theta = 45$ $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}}x_1 - \frac{1}{\sqrt{2}}x_2 \\ \frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 \end{pmatrix}$

Point (x_1, x_2)

$0,0 \to 0,0$

$1,0 \to \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

$(0,1) \to -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

$(1,1) \to (0, \sqrt{2})$

Lets have few more examples

→ $T: \mathbb{R}^2 \to \mathbb{R}^3$ such that

$T(x_1, x_2) = (x_1 + x_2 + 1,\ 2x_1 - 3x_2;\ 9x_1 + 7x_2)$

Lets try $(0,0)$

$T(0,0) = (1, 0, 0)$

Linear map fails as $(0,0)$ doesn't go to $(0,0,0)$.

→ $T: M_2(\mathbb{R}) \to M_2(\mathbb{R})$ such that

$T(A) = I + A$

Zero mx will give here Identity mx.

Here also this fail.

→ $T: \mathbb{R}^2 \to \mathbb{R}^4$ such that

$T(x_1, x_2) = (x_1 + x_2,\ x_1 - x_2,\ 2x_1 + x_2,\ 3x_1 - 4x_2)$

$T(0,0) = (0, 0, 0, 0)$ ✓

$T(v_1) = T(x_1, x_2) = [x_1 + x_2,\ x_1 - x_2,\ 2x_1 + x_2,\ 3x_1 - 4x_2]$ — (A)

$T(v_2) = T[y_1, y_2] = [y_1 + y_2,\ y_1 - y_2,\ 2y_1 + y_2,\ 3y_1 - 4y_2]$ — (B)

L7-12a

$T(v_1+v_2) = T(x_1+y_1, x_2+y_2)]$

$= [x_1+y_1+x_2+y_2, x_1+y_1-x_2-y_2, 2x_1+2y_1+x_2+y_2,$
$3(x_1+3y_1 -4x_1-4y_2)$

$= [x_1+x_2+y_1+y_2, x_1-x_2+y_1-y_2, 2x_1+x_2+2y_1+y_2,$
$3x_1-4x_1+3y_1-4y_2]$

(A) + (B)

$T(v_1)+T(v_2)$
$= [x_1+x_2+y_1+y_2, x_1-x_2+y_1-y_2, 2x_1+x_2+2y_1+y_1,$
$3x_1-4x_2+3y_1-4y_2)$

$\rightarrow T(\alpha x_1, \alpha x_2) = (\alpha x_1+\alpha x_2, \alpha x_1-\alpha x_2, 2\alpha x_1+2\alpha x_2,$
$3\alpha x_1 - 4\alpha x_2)$

$= \alpha(x_1+x_2, x_1-x_2, 2x_1+2x_2,$
$3x_1-4x_2)$

$\alpha T(v)$

$\therefore$ It is linear mapping

L7-12b

$\rightarrow \mathbb{R}^2 \rightarrow \mathbb{R}^2$ s.t.

$$T(x_1, x_2) = (x_2 - x_1, 4x_2^2 - x_1^2)$$

$$T(0,0) = (0,0)$$

$$T(v_1) = T(x_1, x_2) = (x_2 - x_1, 4x_2^2 - x_1^2)$$

$$T(v_2) = T(y_1, y_2) = (y_2 - y_1, 4y_2^2 - y_1^2)$$

$$T(v_1 + v_2) = T(x_1 + y_1, x_2 + y_2)$$

$$= [x_2 + y_2 - x_1 - y_1, 4(x_2 + y_2)^2 - (x_1 + y_1)^2]$$

$$= (x_2 + y_2 - x_1 - y_1, 4x_2^2 + 4y_2^2 + 8x_2y_2 - x_1^2 - y_1^2 - 2x_1y_1]$$

$$\neq T(v_1) + T(v_2)$$

$\therefore$ T is not a linear mapping.

$\Rightarrow M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ s.t. $T(A) = A^T$

It is linear mapping